



Configuration and Data (Sheets) Management in EDS Architecture

1. Purpose

This section defines how **content**, **AEM Sheets (tabular data)**, and **configuration** are handled in the target architecture where:

- **AEM** is the **authoring source**
- **Edge Delivery Services (EDS)** is the **delivery layer**
- the solution uses a **repoless EDS setup**

The objective is to make explicit:

- where authored content lives
- where sheets fit in the architecture
- where configuration is managed
- how pages, blocks, sheets, and delivery work together
- how these elements are organized and governed

This section addresses the stakeholder concern that the solution design must clearly identify where **configurations** and **sheets** are addressed as part of the overall architecture.

2. Architectural Overview

In this solution, AEM remains the **system of record for authored content**, while EDS provides the **delivery tier**. Content authors work in AEM, and EDS delivers the final website experience through its edge-based delivery model.

Sheets are used as a **supporting data model** for reusable **tabular datasets** that are better managed as rows and columns than as page-authored fields.

Because this is a **repoless setup**, **site-level configuration and delivery configuration are not treated as normal authored content**. They are managed through the **EDS admin/configuration layer** and associated runtime mechanisms.

Sources:

- [Authoring with AEM for Edge Delivery Services](#)
- [Using Edge Delivery Services with AEM](#)
- [Path mapping for AEM authoring as your content source](#)
- [Repoless AEM Adaptive Forms Sites — Setup & Management Guide](#)

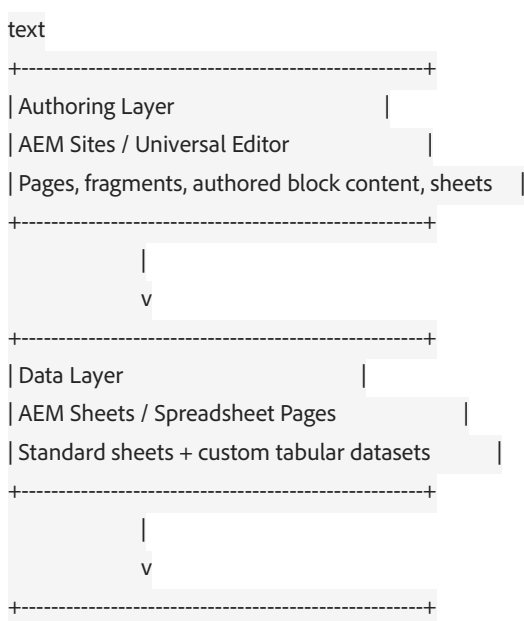
3. Layered Architecture View

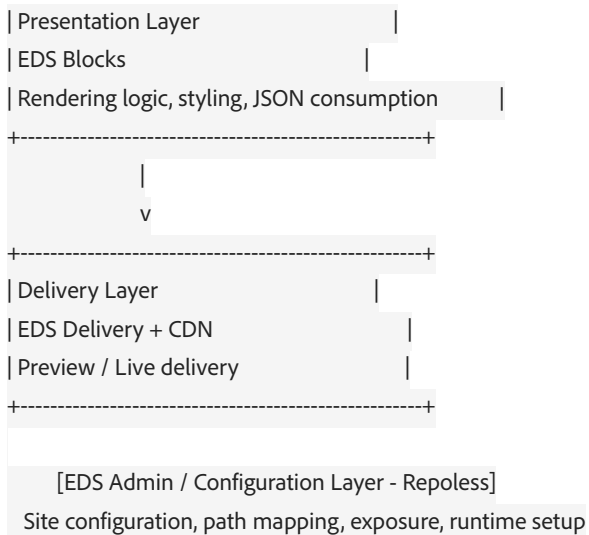
The solution should be understood using the following layered model.

3.1 Logical layers

- 1. Authoring Layer (AEM)**
Manages page content, authored block content, fragments, and sheets.
- 2. Data Layer (Sheets)**
Holds reusable tabular datasets that must be exposed as JSON.
- 3. Presentation Layer (EDS Blocks)**
Implements block rendering, styling, and data consumption.
- 4. Delivery Layer (EDS / CDN)**
Delivers the site through Edge Delivery Services.
- 5. EDS Admin / Configuration Layer**
Manages repoless site configuration, path mapping, resource exposure, and runtime delivery setup.

3.2 Layered architecture diagram





This model is important because it shows that **sheets are part of the content/data layer**, whereas **repoless configuration is part of the EDS admin/configuration layer**.

Sources:

- [Introduction to the Architecture of Adobe Experience Manager as a Cloud Service](#)
- [Using Edge Delivery Services with AEM](#)

4. AEM as Authoring Source in EDS

AEM is used as the **authoring and content management platform** for the solution.

4.1 What is authored in AEM

The following are authored and managed in AEM:

- pages
- page content and page structure
- authored block instances
- Experience Fragments, where applicable
- Content Fragments, where applicable
- sheet pages used for tabular data
- standard EDS sheet types such as metadata or redirects, where required

4.2 Role of AEM in the architecture

AEM provides:

- content authoring
- editorial governance
- workflows and approval processes
- persistence of page and sheet content
- publish events that enable content to flow to EDS

In this model, AEM remains the **content source**, while EDS is the **delivery platform**.

Sources:

- [Authoring with AEM for Edge Delivery Services](#)
 - [Methods to Author Content in AEM](#)
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5. AEM Sheets / Spreadsheets in EDS

5.1 What sheets are

In this architecture, **sheets are AEM spreadsheet pages used to maintain tabular data**. AEM translates these spreadsheets into **JSON**, which can then be consumed by the site or by EDS blocks.

Adobe positions spreadsheets for use cases such as:

- Metadata
- Headers
- Redirects
- Placeholders
- Taxonomy
- Configuration-style tabular datasets
- Custom project-specific mappings and lookup tables

5.2 When sheets are used

Sheets are used when the required information is:

- tabular
- reusable
- shared across pages or blocks

- author-manageable
- appropriate to expose as JSON

Typical examples include:

- reusable lookup values
- block-specific mappings
- row/column business data
- bounded rule tables

5.3 How sheets are used by blocks

Sheets are not used as end-user pages. Instead:

1. a sheet is authored in AEM
2. the sheet is published and exposed through the EDS path/configuration model
3. the sheet becomes available as JSON
4. a block consumes that JSON and renders the required output

This makes sheets suitable for **data-driven blocks**.

5.4 Standard sheet types

AEM provides standard sheet templates for EDS-related use cases, including:

- Metadata
- Headers
- Redirects
- Placeholders
- Taxonomy
- Configuration
- Spreadsheet (generic/custom)

Sources:

- [Managing tabular data with AEM authoring as your content source](#)
- [Spreadsheets and JSON](#)

6. Explicit Differentiation: Sheets vs Configuration

A key clarification in this solution is that **sheets** and **configuration** serve different purposes.

6.1 Sheet-based data

Sheets are used for:

- business-managed tabular data
- reusable mappings
- row/column datasets
- data that blocks or frontend logic consume as JSON

6.2 Configuration

Configuration is used for:

- site setup
- delivery behavior
- resource exposure
- path mapping
- block definitions and behavior rules
- environment-specific delivery settings

6.3 Comparison table

Aspect	Sheets	Configuration
Purpose	Business/data-oriented tabular content	System/delivery-oriented setup and behavior
Managed by	Content/business teams with governance	Architecture, development, platform/admin teams
Stored in	AEM spreadsheet pages	EDS admin/configuration layer and project/block configuration
Output	JSON data for consumption	Delivery/runtime behavior and exposure rules

Aspect	Sheets	Configuration
Typical examples	redirects, metadata, lookup tables, table data, block mappings	path mapping, site linkage, block configuration, runtime delivery settings

6.4 Repoless clarification

In the **repoless EDS setup**, **site-level configuration is not treated as author-managed sheet content**. It is handled through the **EDS admin/configuration layer**. Sheets are used only where the solution intentionally requires **tabular data exposed as JSON**.

Sources:

- [Managing tabular data with AEM authoring as your content source](#)
- [Path mapping for AEM authoring as your content source](#)
- [Repoless AEM Adaptive Forms Sites — Setup & Management Guide](#)

7. Configuration Management in Repoless Architecture

For this solution, configuration exists in **three distinct layers**.

7.1 AEM layer

AEM stores:

- page content
- authored block content
- sheet pages
- fragments where applicable

This is the **content and data authoring layer**.

7.2 EDS blocks layer

EDS blocks define:

- frontend rendering behavior
- block styling
- how authored content is displayed

- how sheet JSON is consumed

This is the **presentation and behavior layer**.

7.3 EDS Admin / Configuration layer

The repoless EDS admin/configuration layer manages:

- site configuration
- content-source linkage
- path mapping
- exposure of pages, assets, and sheets
- runtime and delivery configuration

This is the **delivery and runtime configuration layer**.

7.4 Key principle

In this solution:

- **AEM manages content and sheets**
- **EDS blocks manage rendering behavior**
- **EDS admin/configuration services manage repoless site configuration**
- **path mapping and exposure are handled outside normal AEM content**

Sources:

- [Content modeling for AEM authoring projects](#)
- [Using Edge Delivery Services with AEM](#)
- [Path mapping for AEM authoring as your content source](#)

8. Content Flow and Architecture Integration

8.1 End-to-end flow

text

Author

-> AEM Sites / Universal Editor

-> AEM Page Content or AEM Sheet

-> Publish

-> AEM renders semantic HTML or exposes sheet as JSON
-> EDS ingests content
-> EDS / CDN delivers site
-> End User

8.2 Relationship between pages, blocks, and sheets

- **Pages** define the authored experience and page composition
- **Blocks** define presentation and behavior
- **Sheets** provide reusable tabular data where blocks need shared JSON-backed data

text

Page

- ☒ Block A -> uses page-authored content only
- ☒ Block B -> uses page-authored content + sheet JSON
- ☒ Block C -> uses sheet JSON for reusable tabular data

This is the governing relationship for the solution:

- **pages control placement**
- **blocks control rendering**
- **sheets provide reusable data**
- **EDS config controls exposure and delivery**

Sources:

- [Authoring with AEM for Edge Delivery Services](#)
- [Content modeling for AEM authoring projects](#)
- [Managing tabular data with AEM authoring as your content source](#)

9. Sheet Storage and Organization

Sheets should be stored in the AEM Sites content structure using dedicated folders that separate:

- standard sheets
- block-specific sheets
- business-owned data sheets

9.1 Recommended structure

```
text
/content/<site-root>/
  /pages/
    /en/
    /fr/
  /configuration/
    /standard/
      metadata
      headers
      redirects
      placeholders
      taxonomy
      configuration
    /blocks/
      /<block-name>/
        <block-specific-sheet>
  /data/
    /tables/
      <business-table-sheet>
```

9.2 Organization principle

- **/configuration/standard**
For standard EDS sheet types
- **/configuration/blocks**
For block-specific reusable tabular sheets
- **/data/tables**
For business-owned row/column datasets

This keeps page content separate from reusable tabular resources and makes ownership and maintenance clearer.

Sources:

- [Managing tabular data with AEM authoring as your content source](#)

10. Application in Current Solution

The current solution applies the folder structure as follows.

10.1 **/configuration/standard**

Used for standard AEM sheet types where required, such as:

- metadata
- headers

- redirects
- placeholders
- taxonomy
- configuration

These support standard site-wide tabular artifacts.

10.2 /configuration/blocks

Used for **block-specific reusable sheets**, where a specific feature or block needs tabular data maintained outside page-level authoring.

Examples to be finalized per project:

- <block-specific-config-sheet>
- <rules-sheet>
- <lookup-sheet>

10.3 /data/tables

Used for **business-owned tabular datasets** rendered by one or more blocks.

Examples to be finalized per project:

- <table-data-sheet>
- <comparison-sheet>
- <catalog-sheet>

This makes the intended solution mapping clear:

- **standard sheets** -> /configuration/standard
- **block-specific reusable sheets** -> /configuration/blocks
- **business data tables** -> /data/tables

11. Publication and Path Mapping

Sheets are not consumed directly from AEM authoring paths. They must be exposed through the **EDS path/configuration model** so they are available as public JSON resources.

11.1 Typical endpoint patterns

- `metadata` -> `/metadata.json`
- `headers` -> `/.helix/headers.json`
- `configuration` -> `/.helix/config.json`
- `<block-specific-sheet>` -> `/config/<name>.json`
- `<table-data-sheet>` -> `/data/<name>.json`

11.2 Important clarification

This means:

- sheets live in AEM
- path mapping and exposure are controlled outside page authoring
- blocks consume published JSON endpoints, not AEM authoring URLs directly

In the repoless model, this exposure is part of the **EDS admin/configuration layer**.

Sources:

- [Path mapping for AEM authoring as your content source](#)
 - [Managing tabular data with AEM authoring as your content source](#)
-

12. Environment and Deployment Considerations

12.1 Sheets across environments

Sheets are authored in AEM and therefore follow the same content lifecycle as other AEM-managed content.

This means:

- lower environments may contain draft or test sheet data
- production contains approved business data
- preview/publish lifecycle applies to sheets as it does to other content resources

Accordingly, sheet values may differ between QA and Production based on:

- approval state
- business readiness
- environment-specific test data

12.2 Configuration across environments in repoless setup

In the repoless architecture:

- **sheet content** is managed as authored content
- **delivery configuration** is managed through the EDS admin/configuration layer
- **path mapping and endpoint exposure** are managed per environment outside normal AEM page content

This distinction is important:

- **sheet content is content-managed**
- **path mapping and delivery exposure are environment-managed**

Sources:

- [Path mapping for AEM authoring as your content source](#)
 - [Repoless AEM Adaptive Forms Sites — Setup & Management Guide](#)
 - [Crosswalk - Documentation - General \(title tbd\)](#)
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13. Naming Conventions

A consistent naming convention should be used for custom sheets.

13.1 Naming pattern

text
<type>-<domain>-<purpose>

13.2 Examples

- config-media-brightcove
- config-forms-rules
- data-products-comparison

13.3 Rules

- lowercase only
- hyphen-separated
- no spaces or special characters

- names should clearly indicate purpose
 - **config** and **data** sheets should remain distinguishable
-

14. Governance and Ownership

14.1 Ownership

Artifact	Owner
Page content	Content authors / content team
Standard sheets	Content operations / site admins
Block-specific sheets	Business owner with development oversight
Business data sheets	Business/content owners
EDS blocks and rendering logic	Development team
Path mapping and repoless site configuration	Architecture / development lead / platform team

14.2 Governance principles

- use sheets only for approved tabular use cases
- separate standard sheets from custom project sheets
- group custom sheets by consuming block or business domain
- treat published JSON endpoints as stable contracts
- require review for new sheet types or schema changes
- keep site/delivery configuration outside normal authored content

Sources:

- [Managing tabular data with AEM authoring as your content source](#)
 - [Spreadsheets and JSON](#)
-

15. Conclusion

The target architecture now clearly defines where **configurations** and **sheets** fit in a **repoless EDS implementation**

using AEM as the authoring source.

Specifically:

- **AEM** is the authoring source for pages, authored content, fragments, and sheets
- **AEM Sheets** are used for reusable **tabular datasets** that must be exposed as JSON
- **EDS blocks** are the presentation layer that consume authored content and, where needed, sheet data
- **EDS Admin / configuration services** manage **repoless site configuration, path mapping, resource exposure, and delivery/runtime behavior**

As a result, both **configuration** and **sheet-based data** are now explicitly placed in the overall solution architecture rather than being treated as implicit implementation details. This directly addresses the stakeholder concern regarding where configurations and sheets fit within the solution design.

References

- [Authoring with AEM for Edge Delivery Services](#)
- [Using Edge Delivery Services with AEM](#)
- [Managing tabular data with AEM authoring as your content source](#)
- [Spreadsheets and JSON](#)
- [Path mapping for AEM authoring as your content source](#)
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- [Crosswalk - Documentation - General \(title tbd\)](#)
- [Crosswalk - Documentation - Content Modeling](#)

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